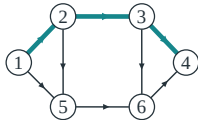


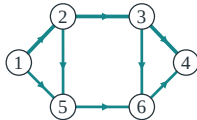
Constraint-like limit



$\varepsilon \rightarrow 0$

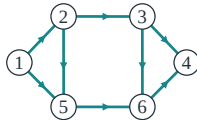
Constrained graph LP

Entropic graph solve



finite ε

Penalty-like limit



$\varepsilon \rightarrow \infty$

Clipped active-set price

Same graph and budgets; vary ε

SafeOT-Dual: one price rule

$$\lambda = \underbrace{\text{filtered clip}(\lambda_\varepsilon^*, 0, c)}_{\text{graph-solved price}} + \underbrace{\beta}_{\text{cost-feedback price}}$$

$$A_{\text{update}} = (A_r - \sum_k \lambda_k A_{c_k}) / (1 + \sum_k \lambda_k)$$

$c = 0$

Fixed β : fixed penalty

Adaptive β : Lagrangian